

Simon Spivey

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EDUCATION

Carnegie Mellon University

Master of Science in Electrical & Computer Engineering

GPA: 4.0 / 4.0

Pittsburgh, PA

December 2024

Olivet Nazarene University

Bachelor of Engineering: Software Engineering, Electrical & Computer Engineering

GPA: 3.73 / 4.0

Bourbonnais, IL

May 2021

SKILLS

Programming Languages:

Lua, C, JS/TS, C#, C++, Python

Database:

MongoDB, SQL, Redis, Elasticsearch,

Tools & Technologies:

Docker, Git, SSH, Hyper-V, Azure DevOps, GitHub Actions, nginx

PROFESSIONAL EXPERIENCE

Adatafy™

Remote

Solutions Architect Consultant

September 2023 – Present

- Created a synchronization system that in real-time syncs work order, location, and event data across multiple systems, enhancing operational efficiency, data accuracy, and saving money on licensing.
- Engineered a debugger that enables developers to examine call stacks, variables, and breakpoints in real-time distributed Lua code, streamlining and reducing the debugging headaches.
- Launched a website that allows employees to search, input, and research products and tools across the global Emerson network, avoiding potential redundant work and keeping sales teams informed.
- Created a reporting tool that generates a report based on the time selected to aggregate 100k+ pieces of version-control data to satisfy pharmaceutical quality control regulations.

Novaspect Inc.

Schaumburg, IL

IoT Engineer

June 2021 – August 2023

- Engineered an air compressor testing system using PLC, SCADA, and BaaS technologies to automate the testing process from assembly line to specification sheet to meet new DoE standards.
- Developed a system that automates master batch production and generates KPIs and BI data for each batch, allowing remote experts to safely modify and analyze production lines.
- Mocked a system that could scale to handle 100's of Petabytes of data within an air-gapped cloud-like ecosystem, which received government funding for the initial phase.

PROJECTS

Carnegie Mellon University

Pittsburgh, PA

- Designing Sustainable Cloud Systems – Designed a cloud service focused on availability in environments where stability and bandwidth are limited. This led to active research within CMU.
- Fault-Tolerant Distributed Systems – Lead a team of 4 students to build a fault-tolerant distributed system that spanned ~8 microservices (~20 with replication) and could handle (n-1) crashes.
- Decentralized Distributed RC Car – Created a RTOS for two ARM cortex programmed to control zones of the car and can be a redundant pair in failure. Both receive driving signals from a remote cockpit.